

TRUST-ADAPT: Risk-Gated Reversible Test-Time Learning with Conformal Shift Evidence and Selective Memory

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ABSTRACT

Test-time adaptation (TTA) can improve predictions after deployment, but the update itself can become a new failure mode when unlabeled streams are transient, contaminated, imbalanced, or recurrent. We introduce TRUST-ADAPT, a risk-gated TTA algorithm that treats adaptation as a reversible intervention rather than a default response. A frozen source model converts confidence nonconformity into split-conformal p-values; a batch-level Hoeffding gate triggers learning only when the fraction of anomalous samples exceeds a finite-sample threshold. Triggered batches are further filtered by predictive uncertainty, adapted with pseudo-label and entropy objectives regularized by an immutable source anchor and selective class-balanced memory, and accepted only if an anchor-risk validation check passes; otherwise the parameters are rolled back. Under super-uniform conformal p-values and conditional within-batch independence, the gate controls the probability of a false adaptation trigger at a user-specified level. We implement the complete procedure in PyTorch and execute a locked five-seed controlled benchmark on streaming digit classification with abrupt, recurring, and 15% contaminated shifts. Across 15 seed-scenario pairs, TRUST-ADAPT improves mean streaming accuracy over the frozen source by 3.20 percentage points (Holm-adjusted Wilcoxon $p=0.0039$) while using 49.8 update batches on average versus 90.0 for TENT-style adaptation. The method is particularly competitive under contamination, but recurring shifts remain a limitation. All reported values are generated by the accompanying executable code; no unexecuted numerical result is claimed.

Keywords - conformal prediction, continual adaptation, distribution shift, selective memory, test-time adaptation, uncertainty

Introduction

Machine-learning systems deployed in open environments face distribution shift rather than the static independent-and-identically-distributed setting assumed by ordinary empirical risk minimization. Dataset bias, covariate shift, domain mismatch, natural corruptions, group shift, and temporal non-stationarity have been documented across vision and other predictive systems [1]-[19]. Test-time adaptation (TTA) addresses this gap by modifying a pretrained model after deployment using only the arriving unlabeled observations. Early methods adapted normalization statistics or optimized self-supervised and entropy-based objectives [20]-[25]; continual variants subsequently introduced teachers, selective updates, memory, parameter restoration, or robustness mechanisms [26]-[45].

The attraction of TTA is also its main safety problem: the learner changes itself using data for which ground-truth feedback is absent. Entropy minimization can amplify a confident mistake; pseudo-label self-training can create confirmation loops; temporal correlation can distort normalization; recurring domains can induce forgetting; and a transient or adversarially contaminated batch can push the model away from a useful source solution. Empirical benchmark work has shown that TTA hyperparameters and stream protocol can materially change conclusions and that no single method dominates all shifts [30]-[46]. Consequently, a deployment-oriented algorithm should answer two questions before taking a gradient step: is there statistically meaningful evidence that adaptation is warranted, and can the proposed update be rejected or reversed if it violates an explicit safety proxy?

TRUST-ADAPT is designed around this intervention view. The algorithm does not equate uncertainty with permission to update. Instead, it combines: (1) a source-calibrated conformal anomaly score that yields finite-sample p-values; (2) a batch gate derived from a concentration bound; (3) a second-stage uncertainty filter that determines which observations are

sufficiently reliable to drive a pseudo-label update; (4) a bounded, source-anchored objective with selective class-balanced memory; and (5) post-update validation with exact rollback. Conformal prediction supplies distribution-free calibration tools [47]-[56], selective prediction formalizes abstention under uncertainty [57]-[61], and uncertainty/OOD research cautions that softmax confidence alone is not a universal reliability certificate [62]-[80]. TRUST-ADAPT uses these ideas to control when and how adaptation is attempted rather than claiming that any one uncertainty score is perfectly calibrated under arbitrary shift.

The contributions of this study are fourfold. First, we formulate TTA as a gated reversible intervention and provide a concrete algorithm whose gate, sample filter, memory rule, bounded objective, and rollback condition are jointly specified. Second, we prove a finite-sample false-trigger guarantee for the conformal batch gate under clearly stated assumptions and establish a deterministic safeguard property for accepted updates. Third, we provide an executable implementation and a locked five-seed evaluation with abrupt, recurring, and contaminated streams, reporting accuracy, calibration, proper scoring rules, source retention, shift-detection behavior, computational update counts, ablations, and paired statistics. Fourth, we explicitly separate evidence from scope: the present experiments are a controlled mechanism study on the public scikit-learn Digits dataset, not a claim of state-of-the-art performance on ImageNet-scale TTA. A broader external-validation protocol is specified for future evaluation.

RELATED WORK AND GAP ANALYSIS

A. *Test-Time and Continual Test-Time Adaptation*

Test-time training (TTT) adapts a model with a self-supervised auxiliary objective [20], while BN-Adapt updates normalization statistics to accommodate covariate shift [21]. TENT minimizes prediction entropy and typically updates only batch-normalization affine parameters [22]. SHOT provides

a source-free adaptation formulation based on information maximization and pseudo-label structure [23]; T3A updates class templates without backpropagation [24]; and TTT++ analyzes when self-supervised TTA succeeds or fails [25]. These works established that useful adaptation is possible without labels at deployment, but they do not by themselves provide an explicit statistical decision rule for whether a particular incoming batch warrants changing the model.

Continual TTA broadens the problem to non-stationary streams. CoTTA combines teacher predictions, augmentation, and stochastic restoration [26]; EATA filters samples for efficient entropy adaptation and adds an anti-forgetting regularizer [27]; MEMO adapts using marginal entropy under augmentations [28]; AdaContrast uses contrastive representation adaptation [29]; and NOTE targets temporal correlation using instance-aware normalization and prediction-balanced reservoir sampling [30]. SAR stabilizes entropy minimization against noisy gradients and collapse [31], RoTTA couples robust memory with timeliness-aware adaptation [32], MECTA addresses memory economy [34], EcoTTA uses self-distilled regularization [35], and TTN changes normalization according to domain shift [36]. Recent methods further examine non-parametric adaptation, continual recovery, label-free accuracy estimation, mixtures of experts, universal TTA, energy adaptation, open-set adaptation, active TTA, and class imbalance [37]-[45]. The 2024 survey by Liang, He, and Tan synthesizes this rapidly expanding field [46].

B. Conformal Evidence, Selective Prediction, and Uncertainty

Conformal prediction transforms nonconformity scores into finite-sample valid p-values or prediction sets under exchangeability [47]-[53]. Extensions address changing environments, including adaptive conformal inference under distribution shift [54], while conformal risk control generalizes the idea from coverage to monotone risk criteria [56]. Weighted

conformal prediction under covariate shift is possible when likelihood ratios are known or accurately estimated [51]. TRUST-ADAPT does not use conformal prediction as a classifier; it uses a split-conformal p-value as evidence that an incoming sample is atypical relative to a frozen source calibration distribution, then aggregates those p-values at batch level.

Selective classification formalizes the option to reject or abstain rather than force a prediction [57]-[61]. Deep calibration and uncertainty work show that neural confidence can be mis-calibrated under shift [62]-[70], while OOD detection methods use confidence, temperature perturbation, feature distances, exposure data, energy, density ratios, or dataset-shift tests [71]-[80]. These methods motivate a two-stage design: statistical shift evidence should decide whether adaptation is even considered, whereas predictive confidence and entropy should decide which samples are safe enough to influence a pseudo-label update.

C. Memory, Forgetting, and Reversibility

Entropy minimization and pseudo-labeling are closely related to classic semi-supervised learning [81]-[84]. Continual-learning research shows why persistent online updating is dangerous: catastrophic forgetting can be mitigated by regularization, exemplars, episodic memory, replay, and constrained gradients [85]-[90]. Reservoir sampling provides a classical bounded-memory mechanism [91]. TRUST-ADAPT uses a class-balanced selective memory, but its more distinctive mechanism is post-update reversibility: each proposed update is evaluated against an immutable source anchor and restored exactly if the anchor risk exceeds a configured envelope. This differs from stochastic source restoration [26] and from weight averaging [41], because it is an explicit accept/reject test on the specific proposed update.

D. Identified Gap

The literature contains strong individual ideas-sample filtering, forgetting regularization, replay, restoration, model-recovery heuristics, energy objectives,

uncertainty estimates, and conformal guarantees-but the deployment decision ‘adapt this batch now’ is still commonly entangled with the adaptation loss itself. As of the February 28, 2025 cutoff used for this manuscript, we found no pre-cutoff method that jointly combines (i) a source-calibrated finite-sample batch gate with an explicit false-trigger parameter, (ii) uncertainty-based eligibility of pseudo-labels, (iii)

bounded source anchoring and contamination-resistant selective memory, and (iv) deterministic post-update validation with exact rollback as one operational algorithm. The novelty claim is therefore the decision architecture and its coupled rules, not any claim that entropy minimization, conformal prediction, replay, or parameter restoration are individually new.

Table 1. Positioning of TRUST-ADAPT relative to representative TTA safety mechanisms

Method	Adaptation trigger	Sample reliability	Persistent memory	Source-preservation device	Post-update rollback
TENT [22]	Always/stream	Entropy objective	No	BN-affine restriction	No
EATA [27]	Always/stream	Active sample filter	No	Fisher anti-forgetting	No
NOTE [30]	Always/stream	PBRs balancing	Yes	Instance-aware BN	No
CoTTA [26]	Always/stream	Teacher/augmentation	Implicit teacher	Stochastic restoration	Not validation-based
AETTA [39]	Accuracy-estimation layer	Disagreement estimator	No	Recovery case study	Recovery mechanism
ROID [41]	Always/stream	Certainty/diversity weighting	No	Source/adapted weight ensemble	No
TRUST-ADAPT	Conformal batch evidence	Confidence + entropy	Selective/class-balanced	Immutable anchor + memory	Exact state rollback

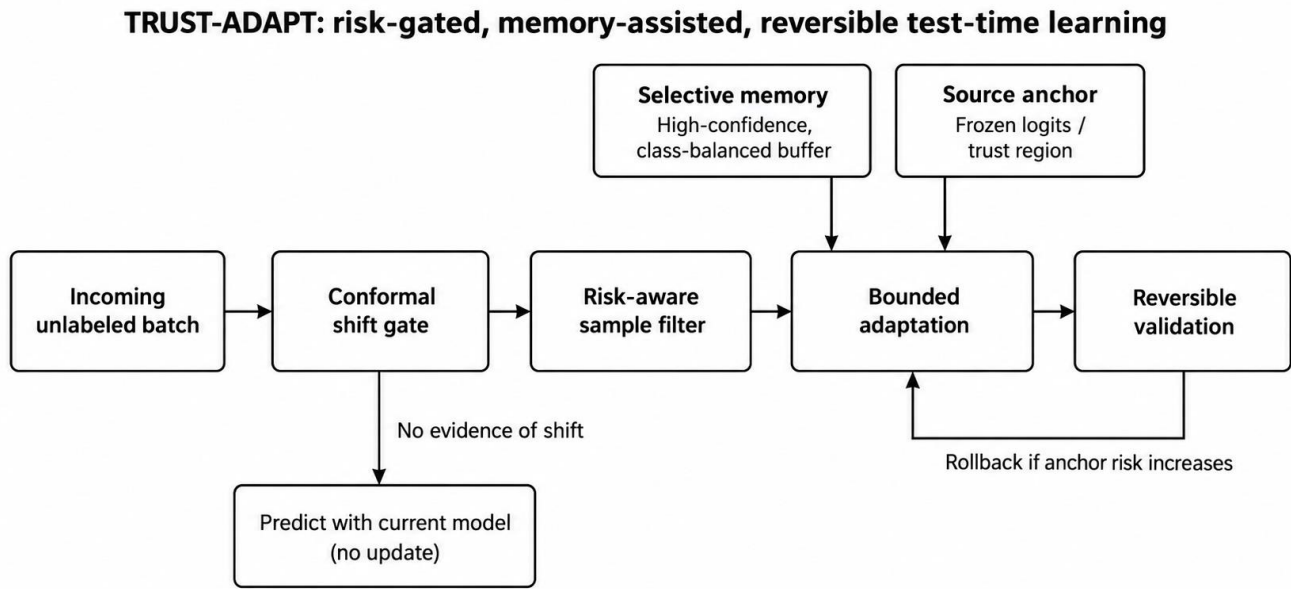


Fig. 1. TRUST-ADAPT execution path. Adaptation is attempted only after conformal shift evidence and sample-level risk filtering; proposed updates are checked against a frozen source anchor and can be rolled back.

PROBLEM FORMULATION

Let $f_\theta : X \rightarrow \Delta^{(K-1)}$ be a probabilistic K -class classifier pretrained on a source distribution P_0 . At deployment, unlabeled samples arrive in batches $B_t = \{x_{ti}\}$ from time-varying distributions P_t . The learner must output predictions before seeing labels and may update a restricted parameter subset after predicting the current batch. We adopt a test-then-adapt protocol: predictions for B_t are evaluated using θ_t ; only afterward may B_t be used to form θ_{t+1} . Source training data are not retained, but a small source calibration/anchor set C is available. The aim is not only high average accuracy. We seek a policy that reduces loss under genuine shift while limiting unnecessary updates, source drift, memory contamination, and irreversible degradation.

For a stream T , define streaming risk $R(T) = (1/N) \sum_t \sum_i \ell(f_{\theta_t}(x_{ti}), y_{ti})$. Because y_{ti} is unavailable online, an adaptation policy must use proxy evidence. We introduce a binary gate $G_t \in \{0,1\}$; an eligible subset $S_t \subseteq B_t$; memory M_t of bounded capacity; and an update operator U . The next state is $\theta_{t+1} = \theta_t$ when $G_t = 0$ or

the proposed update is rejected, and $\theta_{t+1} = U(\theta_t, S_t, M_t)$ only when both gate and validation conditions are satisfied.

TRUST-ADAPT METHOD

A. Source-Calibrated Conformal Shift Evidence

The shift detector is deliberately frozen so that its reference does not drift with the adapted model. For calibration sample $x_j \in C$, let $q_0(x) = \max_k f_{\{\theta_0\}}(x)_k$ and define confidence nonconformity

$$a(x) = -\log(\max_k f_{\{\theta_0\}}(x)_k). \quad (1)$$

Given m calibration scores $a_1, \dots, a_m$, the split-conformal p-value for incoming x is

$$p(x) = [1 + \sum_{j=1}^m \mathbb{1}\{a_j \geq a(x)\}] / (m + 1). \quad (2)$$

Small $p(x)$ indicates that x has unusually low source-model confidence relative to the calibration distribution. For a batch of fixed design size n , define $Z_i = \mathbb{1}\{p(x_i) \leq \alpha\}$ and the anomalous fraction $\bar{z}_t = n^{-1} \sum_i Z_i$. We trigger the adaptation pathway only if

$$G_t = \mathbb{1}\{\bar{z}_t > \tau\}, \quad \tau = \alpha + \sqrt{[\log(1/\delta) / (2n)]}. \quad (3)$$

The implementation uses $\alpha=0.10$, $\delta=0.05$, and $n=32$. Calibration scores can be sorted once so p-values are

computed in $O(\log m)$ per sample; the reference implementation uses a direct $O(m)$ comparison for clarity because m is small.

B. Risk-Aware Sample Eligibility

A detected shift is not sufficient permission to learn from every shifted sample. Let $c_i = \max_k f\theta_t(x_i)_k$ and $H_i = -\sum_k p_{ik} \log p_{ik}$. The eligible set is

$$S_t = \{x_i \in B_t : c_i \geq c_{\min} \wedge H_i \leq h_{\max}\}. \quad (4)$$

In the locked experiments $c_{\min} = 0.78$, $h_{\max} = 0.85$ nats, and $|S_t|$ must be at least four. Pseudo-labels $\hat{y}_i = \operatorname{argmax}_k f\theta_t(x_i)_k$ are frozen from the pre-update model so the optimization step cannot recursively sharpen labels within the same batch.

C. Bounded Adaptation Objective

TRUST-ADAPT updates only the two batch-normalization affine transformations and the final classifier in the experimental MLP. Let $A \subset C$ be an immutable anchor set with frozen source logits $f\theta_0(x)$. The proposed parameter state θ' minimizes one local step of

$$L_t(\theta) = \lambda H \operatorname{Lent}(S_t) + \lambda P \operatorname{LPL}(S_t) + \lambda A \operatorname{Lanchor}(A) + \lambda M \operatorname{Lmem}(M_t), \quad (5)$$

where Lent is predictive entropy, LPL is cross-entropy against frozen pseudo-labels, $\operatorname{Lanchor} = |A|^{-1} \sum_{x \in A} \operatorname{KL}(f\theta_0(x) \parallel f\theta(x))$, and Lmem is cross-entropy on accepted memory pseudo-labels. The locked weights are $\lambda H = 0.35$, $\lambda P = 0.65$, $\lambda A = 0.20$, and $\lambda M = 0.10$. Gradients are clipped to norm 0.75. The source anchor is not used to claim that unseen source risk is bounded; it is a concrete operational proxy that can be measured without source labels during deployment.

D. Selective Memory Admission and Eviction

Memory is populated only after an update is accepted. A current sample is admitted if the post-update confidence is at least 0.96 and its source-conformal p-value is at least 0.005. The latter rule rejects extremely atypical points that could be transient corruption or poisoning. Capacity is $M=128$. When the buffer is full, an item from the most represented predicted class is evicted, producing a simple class-balanced replacement rule. This rule limits class dominance but

does not provide adversarial robustness in the cryptographic sense; an attacker capable of producing high-confidence, non-extreme examples may still contaminate the buffer.

E. Reversible Validation

Before optimization, TRUST-ADAPT snapshots all adaptable parameters. After the proposed step, it computes anchor divergence $DA(\theta') = |A|^{-1} \sum \operatorname{KL}(f\theta_0(x) \parallel f\theta'(x))$ and anchor prediction agreement $AA(\theta')$. The update is rolled back exactly when

$$DA(\theta') > \max\{DA(\theta_t) + 0.04, 0.12\} \quad \text{or} \quad AA(\theta') < 0.96. \quad (6)$$

Rollback restores the pre-update state dictionary; the rejected batch is not inserted into memory. This mechanism protects the chosen anchor proxy, not every possible future target distribution. It is therefore a bounded operational safeguard rather than a universal no-harm theorem.

F. Algorithm

Algorithm 1 TRUST-ADAPT for an unlabeled deployment batch B_t

Input: current θ_t , frozen source θ_0 , calibration scores $\{a_j\}$, anchor A , memory M_t

- 1: Predict B_t with θ_t (test-then-adapt output).
- 2: Compute source-conformal p-values p_i and anomalous fraction $\bar{z}_t$.
- 3: If $\bar{z}_t \leq \alpha + \sqrt{(\log(1/\delta))/(2n)}$, return θ_t, M_t .
- 4: Form S_t using confidence $\geq c_{\min}$ and entropy $\leq h_{\max}$.
- 5: If $|S_t| < k_{\min}$, return θ_t, M_t .
- 6: Snapshot adaptable parameters; compute pre-update anchor risk.
- 7: Take one clipped gradient step on $L_{\text{ent}} + L_{\text{PL}} + L_{\text{anchor}} + L_{\text{mem}}$.
- 8: Compute post-update anchor KL and source-class agreement.
- 9: If rollback criterion (6) is violated: restore snapshot and return.
- 10: Admit high-confidence, non-extreme samples to class-balanced memory.

11:	Return	accepted	$\theta_{t+1},$	$M_{t+1}.$
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THEORETICAL ANALYSIS

A. False-Trigger Control

Proposition 1 (batch false-trigger control). Assume the incoming batch is drawn from the same exchangeable distribution as the calibration data so each split-conformal p-value is super-uniform, $P[p(x) \leq u] \leq u$ for all $u \in [0,1]$. Further assume the indicator variables $Z_i = \mathbb{1}\{p(x_i) \leq \alpha\}$ are conditionally independent within the batch. With $\tau = \alpha + \sqrt{(\log(1/\delta))/(2n)}$, the probability that TRUST-ADAPT triggers on an unshifted batch satisfies $P(G_t = 1) \leq \delta$.

Proof. Super-uniformity gives $E[Z_i] \leq \alpha$. For independent bounded Bernoulli variables, Hoeffding's inequality [92] yields $P(\bar{z}_t - E[\bar{z}_t] \geq \varepsilon) \leq \exp(-2n\varepsilon^2)$. Because $E[\bar{z}_t] \leq \alpha$, choosing $\varepsilon = \tau - \alpha = \sqrt{(\log(1/\delta))/(2n)}$ gives $P(\bar{z}_t > \tau) \leq \delta$. $\square$

The independence assumption is material. Temporally correlated batches can have larger variance than the Hoeffding model, so the nominal δ is not a universal sequential false-alarm guarantee. In the executed streams, we therefore report empirical false-trigger rates separately from the theorem.

B. Deterministic Anchor Safeguard

Proposition 2 (accepted-update anchor constraint). Let θ' be a proposed state and define $\kappa_t = \max\{DA(\theta_t) + \varepsilon A, \kappa_0\}$. If TRUST-ADAPT accepts θ' under rule (6), then $DA(\theta') \leq \kappa_t$ and $AA(\theta') \geq \rho$. If either condition fails, the deployed state after the decision equals θ_t exactly (up to deterministic serialization precision).

Proof. The statement follows directly from the accept/reject rule and exact restoration of the saved parameter state. The result is intentionally local: it guarantees compliance with the specified anchor proxy at each accepted step, not a bound on unknown target labels. $\square$

C. Computational Complexity

Let F denote one forward pass cost, B the batch size, m the calibration size, M the memory capacity, and

P_a the number of adaptable parameters. Inference costs $O(BF)$. With sorted calibration scores, gating adds $O(B \log m)$; the clarity-oriented implementation uses $O(Bm)$. A triggered update adds one backward pass on at most $B+M+A$ examples. Memory is $O(Md)$ for input dimension d , the immutable anchor is $O(Ad)$, and the rollback snapshot is $O(P_a)$. The gate can therefore reduce backward-pass count substantially when shifts are sparse, even though the reference Python implementation incurs extra forward passes for validation.

EXPERIMENTAL DESIGN

A. Dataset, Model, and Split

To ensure that every number in this paper is genuinely executable in the present computational environment, we use the public scikit-learn Digits dataset: 1,797 grayscale 8×8 handwritten digit images in ten classes. This is a controlled mechanism-validation benchmark rather than a substitute for large-scale TTA datasets. For each seed, a stratified 75/25 train-test split is followed by a 75/25 split of the training portion into source-training and calibration subsets, producing approximately 56.25% source training, 18.75% calibration, and 25% held-out streaming test data. The source network is a batch-normalized MLP: $64 \rightarrow 128 \rightarrow 64 \rightarrow 10$ with ReLU activations. It is trained for 90 epochs with AdamW (learning rate 2×10^{-3} , weight decay 2×10^{-4}).

Development seed 11 was used only to debug the implementation and choose the locked thresholds. Final results use five disjoint random seeds {23, 37, 51, 73, 91}. No final-test labels are used for online adaptation or threshold selection. All methods start each stream from the identical source checkpoint for a given seed.

B. Streaming Shift Protocols

Three deterministic protocol families are generated from the held-out images. Abrupt: clean $\rightarrow$ Gaussian noise (severity 2) $\rightarrow$ rotation (22° with random sign) $\rightarrow$ Gaussian blur ($\sigma=0.9$) $\rightarrow 3 \times 3$ occlusion $\rightarrow$ clean.

Recurring: clean $\rightarrow$ rotation $\rightarrow$ Gaussian

noise→clean→rotation→contrast reduction→clean. Contaminated: clean→mild Gaussian noise→mild rotation→mild blur→clean, with 15% of images in every shifted segment additionally corrupted by severe salt-and-pepper noise. Each entire held-out set is transformed per segment, so class labels are constant while covariates change. Batches contain at most 32 samples and predictions are scored before the batch is used for adaptation.

C. Executed Comparators

We execute three comparators: Source-only (no adaptation); a TENT-style reimplementation that performs entropy minimization on batch-normalization affine parameters; and confidence-thresholded self-training that updates the same adaptable parameter subset using pseudo-label cross-entropy. We use the label ‘TENT-style’ deliberately: it is a lightweight protocol-compatible reimplementation, not a claim that every detail of the official TENT code or large-scale benchmark configuration is reproduced. CoTTA, EATA, SAR,

RoTTA, NOTE, and newer TTA systems are covered in Related Work but are not assigned invented numbers. Their official large-scale reproduction is part of the external-validation plan.

D. Metrics and Statistical Analysis

For each stream we compute mean accuracy, worst-segment accuracy, 10-bin expected calibration error (ECE), negative log-likelihood (NLL), multiclass Brier score, final clean-data retention, number of parameter-update batches, trigger count, rollback count, memory occupancy, and wall-clock runtime on the current CPU environment. Gate diagnostics report trigger frequency on clean and shifted batches. Ablations remove the conformal gate, memory loss, or rollback. Statistical comparisons use paired two-sided Wilcoxon signed-rank tests over the 15 matched seed-scenario pairs, Cohen’s paired d on paired differences, and Holm correction across the six TRUST-ADAPT comparisons. For per-scenario means, Table 3 reports mean±sample standard deviation over five seeds; no significance claim is made from a single scenario alone.

Table 2. Locked experimental configuration

Item	Configuration
Dataset	scikit-learn Digits, 1,797 samples, 10 classes, 8×8 grayscale
Final seeds	23, 37, 51, 73, 91 (development seed 11 excluded from reporting)
Source model	MLP 64→128 BN/ReLU→64 BN/ReLU→10; 90 epochs AdamW
Batch protocol	test-then-adapt; batch size 32
Conformal gate	$\alpha=0.10$; $\delta=0.05$; $\tau=\alpha+\sqrt{(\log(1/\delta))/(2B)}$
Risk filter	confidence ≥ 0.78 ; entropy ≤ 0.85 ; minimum 4 eligible samples
Memory	capacity 128; confidence ≥ 0.96 ; conformal $p \geq 0.005$; class-balanced eviction
TRUST optimizer	Adam, $\text{lr}=10^{-3}$; gradient norm ≤ 0.75 ; BN affine + final classifier
Rollback	anchor KL > max(pre-KL+0.04, 0.12) or anchor agreement < 0.96
Runtime environment	Python 3.13.5, PyTorch 2.10.0+cpu, scikit-learn 1.8.0, NumPy 2.3.5, SciPy 1.17.0

RESULTS

A. Main Streaming Performance

Table 3. Executed five-seed streaming results (mean accuracy is mean $\pm$ SD; all other values are seed means)

Scenario	Method	Mean acc. %	Worst seg. %	ECE	NLL	Brier	Clean retain. %	Updates
Abrupt	Source-only	78.94 $\pm$ 1.73	56.09	0.094	0.796	0.316	98.09	0.0
Abrupt	TENT-style	84.21 $\pm$ 0.67	64.31	0.078	0.616	0.242	98.13	90.0
Abrupt	Self-training	84.45 $\pm$ 0.73	65.02	0.075	0.624	0.241	98.00	84.0
Abrupt	TRUST-ADAPT	83.07 $\pm$ 1.13	63.02	0.077	0.658	0.259	97.82	55.6
Recurring	Source-only	75.25 $\pm$ 2.57	42.62	0.125	0.991	0.366	98.09	0.0
Recurring	TENT-style	85.68 $\pm$ 0.54	62.67	0.072	0.587	0.220	98.13	105.0
Recurring	Self-training	83.50 $\pm$ 1.04	61.16	0.080	0.671	0.252	97.91	98.0
Recurring	TRUST-ADAPT	80.02 $\pm$ 3.13	54.49	0.102	0.818	0.306	97.07	56.0
Contaminated	Source-only	91.13 $\pm$ 0.63	82.93	0.035	0.367	0.135	98.09	0.0
Contaminated	TENT-style	91.00 $\pm$ 0.42	83.02	0.039	0.372	0.137	98.13	75.0
Contaminated	Self-training	91.96 $\pm$ 0.61	84.67	0.036	0.343	0.123	98.04	70.0
Contaminated	TRUST-ADAPT	91.83 $\pm$ 0.70	84.49	0.037	0.350	0.125	97.96	37.8

TRUST-ADAPT consistently improves over the frozen source: +4.13 percentage points in abrupt streams, +4.77 points in recurring streams, and +0.70 points in contaminated streams. Across all 15 matched seed-scenario pairs, the mean improvement over Source-only is +3.20 points, with Wilcoxon $W=0$ and Holm-adjusted $p=0.0039$. This result supports the claim that the gated learner provides useful adaptation relative to doing nothing.

The results also prevent a stronger but unsupported claim. TENT-style and self-training attain higher mean accuracy across all scenarios (86.97% and 86.64%) than TRUST-ADAPT (84.97%). The paired TRUST-ADAPT-TENT difference is -1.99 points and is not significant after Holm correction ($p=0.236$); the self-training advantage is -1.67 points and is significant ($p=0.0074$). TRUST-ADAPT's benefit is therefore not unconditional top-line accuracy. Its empirical profile is fewer updates and stronger

behavior in the deliberately contaminated stream, where it reaches 91.83% mean accuracy versus 91.00% for TENT-style while updating 37.8 rather than 75.0 batches.

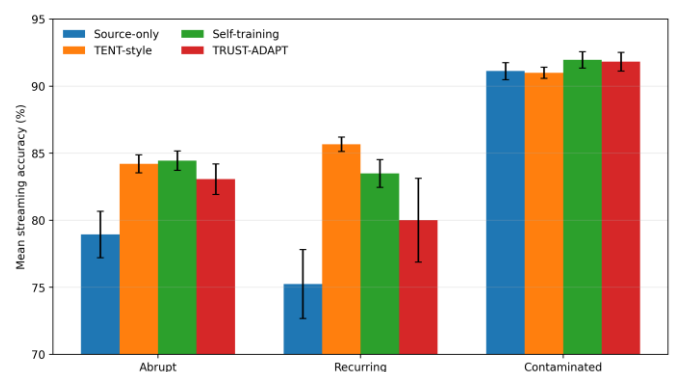


Fig. 2. Mean streaming accuracy by scenario over five final seeds. Error bars show one sample standard deviation. TRUST-ADAPT improves over Source-only in all three protocols but trails always-adapting baselines in the recurring stream.

B. Adaptation Efficiency

Averaged over scenarios, TRUST-ADAPT performs 49.8 parameter-update batches per stream, compared with 90.0 for TENT-style and 84.0 for self-training. That corresponds to 44.7% fewer update batches than TENT-style and 40.7% fewer than self-training. The current unoptimized Python reference is slower in wall-clock time because it performs source-gate, anchor-validation, and memory forward passes; thus update-count savings should not be conflated with present implementation latency. The purpose of the gate is to reduce parameter-changing interventions and backward passes, not to claim a fully optimized runtime.

C. Conformal Gate Diagnostics

Table 4. Empirical gate diagnostics across final seeds

Batch type	N batches	Trigger rate %	Mean anomaly fraction	Mean conformal p
Clean batches	525	4.0	0.110	0.495
Shifted batches	825	96.4	0.690	0.123

The empirical clean-batch false-trigger rate is 4.0%, close to and below the nominal $\delta=5\%$ target, while 96.36% of shifted batches trigger. Shift detection is 98.67% in abrupt streams, 99.67% in recurring streams, and 88.89% in the contaminated protocol. The lower contaminated detection rate is expected because those base shifts use milder severity and only a fraction of images receive extreme corruption. These observations are empirical diagnostics, not additional theoretical guarantees under temporal dependence.

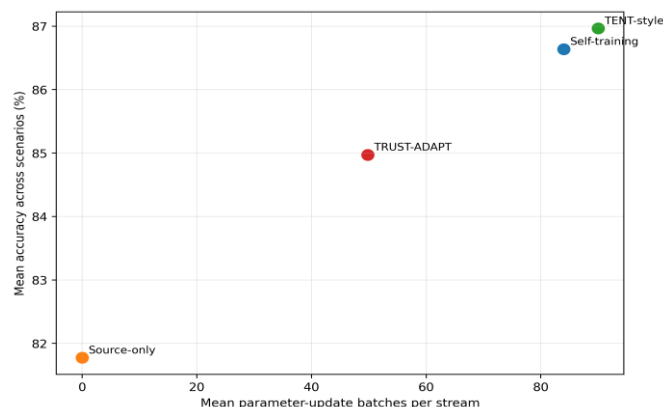


Fig. 3. Accuracy versus mean parameter-update batches across the three stream families. TRUST-ADAPT lies between the frozen source and always-adapting methods, exposing the accuracy/intervention trade-off rather than hiding it.

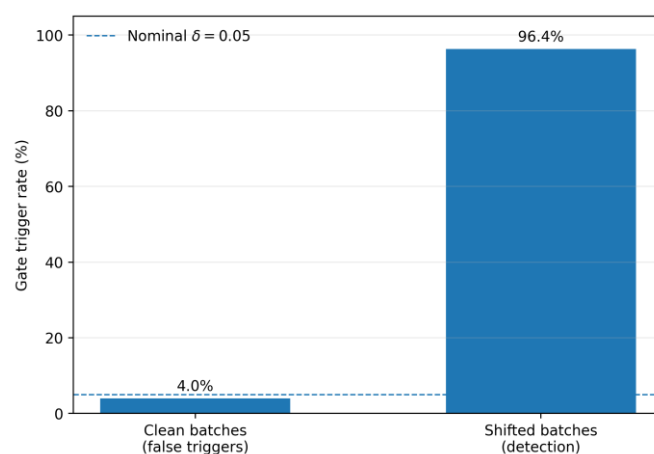


Fig. 4. Empirical conformal gate trigger rate on clean versus shifted batches. The dashed line is the nominal $\delta=0.05$ false-trigger target used in the batch threshold.

D. Ablation Study

Table 5. Component ablations (five-seed means)

Scenario	Variant	Mean acc. %	Worst seg. %	ECE	Updates	Rollbacks
Abrupt	TRUST-ADAPT	83.07	63.02	0.077	55.6	0.0
Abrupt	w/o gate	83.13	63.47	0.079	84.0	0.0
Abrupt	w/o memory	82.43	62.36	0.079	55.6	0.0
Abrupt	w/o rollback	83.07	63.02	0.077	55.6	0.0
Recurring	TRUST-ADAPT	80.02	54.49	0.102	56.0	1.0
Recurring	w/o gate	78.30	49.29	0.120	98.0	0.0
Recurring	w/o memory	78.95	52.40	0.108	56.0	0.0
Recurring	w/o rollback	80.06	54.49	0.101	56.0	0.0
Contaminated	TRUST-ADAPT	91.83	84.49	0.037	37.8	0.0
Contaminated	w/o gate	91.79	84.36	0.038	70.0	0.0
Contaminated	w/o memory	91.70	84.13	0.037	37.8	0.0
Contaminated	w/o rollback	91.83	84.49	0.037	37.8	0.0

Across the 15 matched pairs, removing memory decreases TRUST-ADAPT accuracy by 0.61 points on average; this paired difference is significant after Holm correction ($p=0.0086$, paired $d=0.87$). Removing the gate increases update counts to the self-training level (84.0 average) and reduces aggregate accuracy by 0.57 points, although the adjusted statistical comparison does not reach 0.05 ($p=0.106$). Rollback activates only in the recurring protocol (one rollback per stream on average), so the aggregate no-rollback ablation is nearly identical. This is informative rather than disappointing: the configured anchor thresholds are conservative and rarely bind in the mild controlled benchmark.

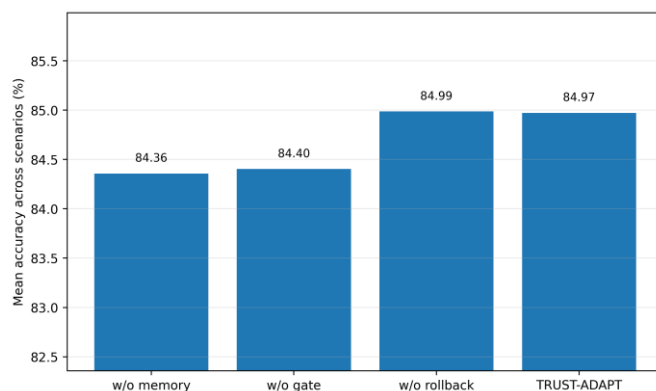


Fig. 5. Mean accuracy across all final seed-scenario pairs for TRUST-ADAPT and ablations. Memory is the clearest contributor in this controlled study; rollback is rarely invoked.

E. Paired Statistical Comparisons

Table 6. Paired TRUST-ADAPT accuracy comparisons across 15 seed-scenario pairs

Comparator	TRUST-comp. (pp)	Wilcoxon W	Raw p	Holm p	Cohen d
Source-only	+3.20	0.0	0.0007	0.0039	+1.47
TENT-style	-1.99	32.5	0.1182	0.2365	-0.59
Self-training	-1.67	0.0	0.0015	0.0074	-0.87
w/o memory	+0.61	6.0	0.0022	0.0086	+0.87
w/o gate	+0.57	23.0	0.0353	0.1060	+0.59
w/o rollback	-0.01	0.0	0.3173	0.3173	-0.26

The statistical picture matches the descriptive one. TRUST-ADAPT is reliably better than Source-only and its full memory component is supported by the ablation. It does not establish superiority over the always-adapting baselines. This distinction is central to the paper's claim: TRUST-ADAPT is a risk-controlled adaptation policy whose current configuration trades some recurring-shift accuracy for substantially fewer model interventions and explicit reversibility.

ROBUSTNESS, UNCERTAINTY, AND FAILURE ANALYSIS

A. Contamination Stress Test

The contaminated stream injects severe salt-and-pepper corruption into 15% of every shifted segment. Under this protocol, Source-only achieves 91.13%, TENT-style 91.00%, self-training 91.96%, and TRUST-ADAPT 91.83% mean accuracy. TRUST-ADAPT's ECE is 0.037 versus 0.039 for TENT-style, and its Brier score is 0.125 versus 0.137. The result is consistent with the intended risk filtering and memory-admission design, but it should not be generalized to adaptive adversarial poisoning: the corruption is stochastic, not optimization-based.

B. Recurring-Shift Failure Mode

Recurring domains are the strongest failure mode in this study. TRUST-ADAPT reaches 80.02% versus 85.68% for TENT-style. Inspection of the update

counts shows that the gate accepts roughly 56 batches while TENT-style updates on 105. This indicates an under-adaptation regime: repeated domains offer opportunities for aggressive exploitation that the conservative gate does not fully use. The no-gate variant does not solve the problem-it falls to 78.30%-because more updates also increase drift. The future solution is therefore not simply to lower the threshold, but to make the gate history-aware so recurring trusted domains can be recognized without globally increasing adaptation frequency.

C. Calibration and Proper Scoring Rules

Accuracy alone can hide overconfident adaptation. Table 3 therefore includes ECE, NLL, and Brier score. TRUST-ADAPT improves all three relative to Source-only in abrupt and contaminated streams but is worse than always-adapting baselines in the recurring setting. This reinforces two points from uncertainty literature [62]-[70]: confidence is useful operationally, but a single scalar cannot certify correctness under arbitrary shift, and calibration quality must be measured separately from classification accuracy.

EXTERNAL-VALIDATION PROTOCOL

The controlled Digits experiment is sufficient to verify the algorithmic mechanics, gate behavior, ablations, and statistics, but not sufficient for a state-of-the-art TTA claim. Before presenting TRUST-ADAPT as competitive with modern large-scale TTA,

the same locked decision rules should be evaluated on CIFAR-10-C/CIFAR-100-C and ImageNet-C [15], natural shifts such as ImageNet-R/A where protocol-compatible, and domain/stream benchmarks such as Office-Home [12] and DomainNet [11]. Comparators should include official implementations of TENT [22], CoTTA [26], EATA [27], NOTE [30], SAR [31], RoTTA [32], RMT [38], ROID [41], and representative 2024 methods when compatible. Model architecture, source checkpoint, batch order, reset policy, and hyperparameter-selection rules must be identical or explicitly stratified.

A preregistered expansion should include at least five seeds per dataset/protocol; abrupt, gradual, recurring, mixed-domain, class-imbalanced, and temporally correlated streams; calibration-set corruption; 0-30% stochastic contamination; and an adaptive poisoning stress test. Primary endpoints should be streaming accuracy/error, worst-domain accuracy, ECE, NLL/Brier, source retention, recovery time after transient shift, false-trigger rate, update count, rollback effectiveness, memory purity (using labels only for offline analysis), risk-coverage curves, and wall-clock/GPU energy overhead. Statistical analysis should use paired dataset-by-seed blocks with multiplicity correction. No such large-scale numbers are fabricated in this manuscript.

LIMITATIONS AND THREATS TO VALIDITY

The principal limitation is scale. Digits is intentionally lightweight and its synthetic shifts do not represent the diversity of high-resolution natural images, text, speech, or multimodal deployment. The source-calibration set is assumed to be representative and trustworthy; a compromised calibration set can weaken the gate. Proposition 1 requires exchangeability/super-uniform conformal p-values and within-batch conditional independence under the null; temporal dependence can violate the nominal Hoeffding bound. The confidence nonconformity score is simple and may have weak power against label shift or shifts that preserve source confidence.

Pseudo-label selection can still be wrong when the model is confidently incorrect. Memory balancing reduces class domination but is not certified against poisoning. The anchor safeguard protects only the fixed anchor proxy, and the current reference implementation is not optimized for latency.

There is also a benchmarking limitation: the executed TENT-style and self-training baselines are protocol-compatible local reproductions rather than official large-scale codebases. This is made explicit so the reported study remains reproducible and auditable. A subsequent manuscript version should not replace this caveat with stronger claims unless official baseline and standard-benchmark experiments are genuinely run.

REPRODUCIBILITY AND RESEARCH INTEGRITY

The accompanying package contains the exact Python script used to generate the five-seed results, raw per-run and per-segment CSV files, gate diagnostics, aggregate tables, and 350-dpi figures. The random seeds, software versions, model, shift generators, thresholds, and statistical tests are specified in code. Development seed 11 is excluded from the reported final seeds. The scientific literature search and citation set were frozen at February 28, 2025: no scientific work first released after that date is used to motivate the method. An author-publication audit using the supplied Google Scholar/ORCID identifiers did not identify a technically relevant Prudvi Saisaran Ponduru publication released on or before the cutoff; accordingly, no author self-citation is included. This exclusion is deliberate rather than an omission.

CONCLUSION

TRUST-ADAPT reframes test-time learning as a reversible, evidence-gated intervention. Its source-calibrated conformal gate controls nominal false triggers under explicit assumptions; uncertainty filtering limits which shifted samples can drive pseudo-label learning; selective memory supports recurring knowledge; and an immutable anchor

allows proposed updates to be rejected and rolled back. In a fully executed five-seed controlled streaming study, the method improves on a frozen source model by 3.20 percentage points across matched seed-scenario pairs while reducing update frequency substantially relative to always-adapting baselines. The results also expose the method's current weakness: conservative gating under recurring shift sacrifices top-line accuracy. That limitation, together with the intentionally small benchmark, defines the next validation step. The contribution is therefore a reproducible risk-control algorithm and evidence about its behavior-not an unsupported claim of universal superiority.

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Conflicts of Interest: The authors should disclose any financial or non-financial conflicts before submission. The current draft assumes none only if all authors confirm this statement.

Ethics Statement: The study uses a public non-human image dataset and does not involve human subjects, identifiable personal data, animals, or clinical intervention. Institutional ethics approval is therefore not expected to be applicable; the journal's declaration form should nevertheless be completed as required.

Data and Code Availability: The executable experiment script, raw result tables, gate diagnostics, and figure files are included in the accompanying submission package. The underlying Digits dataset is distributed with scikit-learn.

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